Appendix E: The Effective Field Theory Limit (QFT Emergence)

E.1 Kaluza-Klein Expansion and Compactification

We define the fundamental 6D bulk action for a massless scalar field $\Phi(x^\mu, y, z)$ as:

$$S_{6D} = -1/2 \int d^4x \int_0^\delta dy\, dz\, \sqrt{(-G)}\, G^{AB}\, \partial_A\Phi\, \partial_B\Phi$$

To recover 4D observable physics, we assume the extra dimensions (y, z) are compactified with characteristic scale δ. We perform a mode expansion of Φ:

$$\Phi(x^\mu, y, z) = \sum_{n,m} \varphi_{nm}(x^\mu)\, \chi_{nm}(y, z)$$

where $\chi_{nm}(y, z)$ are the orthogonal eigenfunctions of the Laplace-Beltrami operator $\nabla^2_{y,z}$, satisfying $\nabla^2\chi_{nm} = -\lambda_{nm}\chi_{nm}$, with eigenvalues λ_{nm} determining the effective mass spectrum.

E.2 Dimensional Reduction and EFT Limit

Substituting this expansion into the 6D action and integrating over the extra coordinates yields the 4D Effective Action:

$$S_{eff} = -1/2 \sum_{n,m} \int d^4x\, [\, \partial^\mu\varphi_{nm}\, \partial_\mu\varphi_{nm} - m_{nm}^{\;2}\, \varphi_{nm}^{\;2}\,]$$

where $m_{nm}^{\;2} = \lambda_{nm}$. At energy scales $E << 1/\delta$, the infinite tower of massive modes $(n, m > 0)$ is suppressed, leaving only the zero-mode φ_{00}. This yields the standard 4D QFT Lagrangian for a massless scalar field.

E.3 Path Integral as a Partition Function

The 6D transition amplitude is defined by the functional integral $Z_{6D} = \int D\Phi \exp(iS_{6D})$. Integrating over the extra-dimensional fluctuations χ_{nm} acts as a Gaussian measure, reducing the 6D propagator G_6 to the 4D Feynman propagator D_F:

$$D_F(x - y) = \langle\, 0 \mid T\{\varphi(x)\, \varphi(y)\} \mid 0\, \rangle = \lim_{\delta\to 0} G_6(x, y; y, z)$$

This formally proves that quantum "virtual particle" interactions in 4D are precisely the harmonic overtones of the 6D interface fabric.